

CareerOS

Complete Product Documentation

"The only platform that tells you exactly why you're not getting interviews — and fixes it."

SRS · PRD · FRD · BRD · Technical Architecture · Go-to-Market

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Internal / Investor Ready

Table of Contents

1.	Executive Summary
2.	Background Story — The Problem We Are Solving
3.	Business Requirements Document (BRD)
4.	Product Requirements Document (PRD)
5.	Functional Requirements Document (FRD)
6.	Software Requirements Specification (SRS)
7.	System Architecture
8.	AI & Machine Learning Design
9.	Database Design
10.	API Design
11.	UI/UX Flow Design
12.	ATS Engine Specification
13.	Job Intelligence Engine
14.	Security & Compliance
15.	Phased Roadmap
16.	Monetization & Business Model
17.	Risk Register
18.	Glossary

1. Executive Summary

Product	CareerOS
Category	AI-Powered Career Intelligence Platform
Primary Market	India — Students, Freshers, Early-Career Professionals
Core Differentiator	Proprietary ATS Engine + Unified Career Profile + Licensed Job Intelligence
Revenue Model	Freemium SaaS · B2B College Licensing · Recruiter Database Access
TAM	\$12B+ (Resume Builder + ATS + Job Intelligence combined)

1.1 Vision

CareerOS is the career operating system for the next billion job seekers. It combines AI-powered resume intelligence, the most accurate ATS simulation engine outside of enterprise HR software, and a legally compliant job intelligence layer — all unified through a single persistent career profile that grows with the user throughout their professional life.

1.2 The Problem

Every year, millions of qualified candidates in India are rejected not because they lack skills, but because their resumes fail automated screening software before a human ever reads them. Over 75% of resumes submitted to large employers are rejected by ATS software before a recruiter sees them (Harvard Business School / Accenture, "Hidden Workers" report). In India, where applicant volumes exceed 500 per posting, this figure approaches 88%.

1.3 The Solution

CareerOS builds recruiter-grade, ATS-optimized resumes through guided AI flows, scores them against a proprietary multi-system ATS engine trained on 15+ real ATS systems, and matches users to live job opportunities through licensed data APIs — all from one persistent career profile filled once and used everywhere.

1.4 Market Opportunity

Metric	Figure	Source / Notes
Engineering graduates/year in India	1.5 million	AICTE 2024
Total annual job seekers (India)	8M+ needing resume help	Estimated across all degree types

Metric	Figure	Source / Notes
Resume builder SaaS market (global)	\$1.2B, 8.3% CAGR	Grand View Research
ATS software market (global)	\$3.2B	MarketsandMarkets
Combined addressable market	\$12B+	Resume + ATS + Job Intelligence
India recruitment tech growth	14% CAGR	NASSCOM 2024

1.5 Target Users

The Fresher (Arjun) (18–22 yrs): Never written a resume. Using a WhatsApp template. Needs full guidance.

The Switcher (Priya) (25–32 yrs): Functional resume but not optimized for target role. Needs ATS tuning.

The Placement Officer (Dr. Ramesh) (35–50 yrs): Manages 2,000 students. Needs institutional tooling and analytics.

2. Background Story — The Problem We Are Solving

2.1 The Story of Arjun

This is a composite story based on patterns observed across thousands of Indian job seekers. It represents the lived experience of the majority of our target users.

Arjun Mehta graduated with a B.Tech in Computer Science from a Tier-2 engineering college in Nagpur in 2023. His CGPA was 8.2. He had completed two internships, built three projects on GitHub, earned certifications in AWS and Python, and spent four months preparing for campus placements.

He applied to 140 companies over six months. He received 3 interview calls.

Arjun was not unqualified. He was invisible.

His resume — a four-page document his senior had shared via WhatsApp, formatted in Word, saved as a PDF — was being fed into Applicant Tracking Systems at Infosys, TCS, Wipro, Accenture, and dozens of startups. Those systems could not read his two-column layout. His skills section used a table format that most ATS parsers ignored. His keywords did not match the exact strings the ATS was filtering for. His file size triggered auto-rejection at two companies.

Arjun had no idea any of this was happening. Neither did his placement officer.

This is not an edge case. This is the norm.

2.2 The Scale of the Problem

According to research by Harvard Business School and Accenture, over 75% of resumes submitted to large employers are rejected by ATS software before a human reads them. In India, this figure approaches 88%. The problem has four distinct layers:

Format Failures: Most Indian students create resumes using Word templates shared on WhatsApp or Canva. These use tables, text boxes, multi-column layouts, and embedded graphics — all of which ATS parsers misread or ignore. A beautiful resume is often structurally invisible to a machine.

Keyword Mismatches: ATS systems filter resumes by matching keywords against job descriptions. "Developed web applications" when the JD says "built full-stack solutions using React and Node.js" may score zero — despite the candidate being fully qualified. Nobody teaches this.

Section Structure Problems: ATS systems are trained to find specific section headers. "My Projects," "Academic Background," or "Technical Expertise" create sections the parser cannot categorize. Data gets lost.

No Feedback Loop: When rejected by an ATS, candidates receive no score, no reason, no improvement path. They reapply with the same resume. They get rejected again. They conclude they are not good enough. The problem is information asymmetry — not capability.

2.3 Why Existing Solutions Fail

Solution	What It Does	Why It Fails Our Users
Canva / Novoresume / Resume.io	Visual resume design	ATS-hostile formatting, no India context, no scoring engine
Jobscan / Rezi	ATS scoring for US market	Expensive (\$30-50/mo), no resume building, no India ATS coverage
Naukri / LinkedIn / Indeed	Job listings	No resume intelligence, no ATS scoring, no guided building
College Placement Cells	Institutional guidance	Understaffed (1 officer per 800 students), no tech tools

The gap is absolute. Nobody has built a vertically integrated, AI-powered system that handles resume creation, ATS optimization, and intelligent job matching in one product, for Indian and emerging market users, at a student-accessible price point.

3. Business Requirements Document (BRD)

3.1 Business Objectives

- Build a self-sustaining, profitable SaaS platform serving 500,000 registered users and 1,000 B2B institutional clients within 24 months of launch.
- Establish CareerOS as the most trusted ATS scoring engine for Indian job seekers.
- Create India's largest verified, ATS-scored resume database, monetizable via recruiter access.
- Build a defensible data moat through proprietary job-resume matching data.
- Achieve product-market fit in Tier-1 cities within 6 months; expand to Tier-2/3 by month 12.

3.2 Stakeholders

Stakeholder	Role	Key Interest
Students / Freshers	Primary End User	Get shortlisted for interviews
Working Professionals	Secondary End User	Career switch, resume refresh
College Placement Cells	B2B Client	Improve placement rates
HR / Recruiters	Revenue Partner	Access quality pre-screened candidates
Investors	Capital Provider	TAM, growth metrics, monetization path
Product & Engineering Teams	Internal	Feature delivery, quality, uptime

3.3 Business Constraints

- Must be profitable at unit level within 18 months of launch.
- Must comply with India's Digital Personal Data Protection Act (DPDP) 2023 from day one.
- Must comply with GDPR for EU/UK users.
- Job data must be sourced only through legally licensed APIs — zero scraping.
- No dependency on a single AI provider (multi-provider fallback required).
- Mobile-first design (70%+ of Indian internet users are mobile-first).

3.4 Success Metrics (Business KPIs)

Metric	6 Months	12 Months	24 Months
Registered Users	50,000	200,000	500,000
Paid Subscribers	2,500	15,000	60,000

Metric	6 Months	12 Months	24 Months
B2B Institutional Clients	20	150	1,000
Monthly Recurring Revenue	■8L	■50L	■2.5Cr
ATS Scans Performed	100,000	800,000	5M+
Resumes Created	75,000	400,000	2M+
NPS Score	40+	55+	65+

4. Product Requirements Document (PRD)

4.1 Product Vision

CareerOS is the career operating system for the next billion job seekers. It combines AI-powered resume intelligence, the most accurate ATS simulation engine outside of enterprise HR software, and a legally compliant job intelligence layer — all unified through a single persistent career profile that grows with the user throughout their professional life.

4.2 User Personas

Persona 1 — The Fresher (Arjun)

- Age 21, B.Tech CSE, Tier-2 college, Nagpur.
- Pain: Has never written a resume. Does not know what ATS is. Using a WhatsApp-shared Word template.
- Goal: Get shortlisted for campus placement drives at IT companies.
- Device: Android smartphone + occasional laptop.
- Willingness to pay: ₹99–₹299/month.

Persona 2 — The Switcher (Priya)

- Age 27, 3 years at a mid-size company, targeting a product company role.
- Pain: Resume is functional but not optimized for product roles. Keeps failing JD-matching.
- Goal: Tailor resume to specific JDs, find the right openings.
- Device: MacBook + iPhone.
- Willingness to pay: ₹499–₹799/month.

Persona 3 — The Placement Officer (Dr. Ramesh)

- Age 45, manages placements for 2,000 students at a private engineering college.
- Pain: Students submit poor resumes at the last minute. Companies give feedback that resumes are weak.
- Goal: Standardize resume quality across the college, improve shortlisting rates.
- Device: Desktop, institutional email, budget-conscious.
- Willingness to pay: ₹1–2L/year for college license.

4.3 Core Product Modules

Module 1: Career Profile Engine

The foundational layer. A single persistent data structure capturing all career-relevant information. This profile powers every other module — resume generation, ATS scoring, and job matching all read from the same source.

Module 2: AI Resume Builder

Guided, conversational resume creation. The AI asks the right questions, understands context (fresher vs. experienced, domain, target role), generates content, and produces ATS-optimized formatted resumes across multiple templates.

Module 3: Proprietary ATS Engine

CareerOS's core differentiator. A multi-system ATS simulation engine that scores resumes against the parsing logic of 15+ real ATS systems, weighted by prevalence in Indian and global markets.

Module 4: Job Intelligence Layer

Powered by licensed job data APIs. Matches the user's career profile against live job listings, scores compatibility, and surfaces relevant opportunities across local, national, and global markets.

Module 5: Analytics Dashboard

Per-user career health dashboard: ATS scores over time, keyword gap analysis, application tracking, market intelligence (skill trends, salary bands).

4.4 Feature Priority Matrix

Feature	Priority	Phase	Complexity	Impact
Career profile onboarding	P0	1	Medium	Critical
AI resume builder (guided)	P0	1	High	Critical
6+ resume templates	P0	1	Low	High
ATS-safe PDF export	P0	1	Medium	Critical
ATS score engine	P0	1	Very High	Critical
Resume upload + score	P0	1	High	Critical
JD-based scoring	P1	1	High	High
Job matching (licensed APIs)	P1	2	High	High

Feature	Priority	Phase	Complexity	Impact
Compatibility scoring	P1	2	Medium	High
Analytics dashboard	P1	2	Medium	Medium
B2B institutional portal	P2	3	High	High
Recruiter access module	P2	3	Medium	High
Mobile app (native)	P2	3	High	High
Cover letter generator	P2	3	Medium	Medium

5. Functional Requirements Document (FRD)

5.1 Module 1 — Career Profile Engine

FR-CP-001: Multi-step onboarding wizard collecting personal details, professional status (student/fresher/experienced/career break), degree/college, target role/industry, and location preference.

FR-CP-002: Unlimited work experience entries: company, title, dates, employment type, location, and up to 8 responsibility bullets each.

FR-CP-003: Unlimited education entries: institution, degree, field, graduation year, CGPA/percentage, relevant coursework.

FR-CP-004: Structured sections: Technical Skills, Soft Skills, Certifications (with issuer, date, credential ID, URL), Projects (title, description, tech stack, GitHub/live URLs), Achievements, Publications, Languages, Volunteer Work, Professional Summary.

FR-CP-005: Career profile persists across sessions; serves as single source of truth for resume generation, ATS scoring, and job matching.

FR-CP-006: LinkedIn OAuth2 import flow to pre-populate career profile from LinkedIn data.

5.2 Module 2 — AI Resume Builder

FR-RB-001: Two modes: (a) Guided AI Mode — conversational LLM-powered step-by-step flow; (b) Direct Edit Mode — structured form with live preview.

FR-RB-002: Guided AI Mode dynamically adjusts questions by professional status. Freshers are not asked for management experience. Experienced users are prompted to quantify achievements.

FR-RB-003: AI content engine generates STAR-format bullet points for work experience with automatic quantification suggestions.

FR-RB-004: Minimum 8 templates: single-column (no photo / with photo), two-column (no photo / with photo), minimal/academic, executive, fresher-optimized, technical/developer.

FR-RB-005: Each template displays ATS compatibility rating clearly.

FR-RB-006: Live preview updates in real-time as user edits.

FR-RB-007: PDF export: text-selectable, standard fonts, no text boxes, no layout tables, no floating elements, under 2MB.

FR-RB-008: Multiple resume versions from the same career profile, each targetable to a different role.

FR-RB-009: AI Professional Summary generator: 3–5 sentences tailored to target role, experience level, and top skills.

FR-RB-010: Section reordering via drag-and-drop.

5.3 Module 3 — ATS Engine

FR-ATS-001: Analyze resumes across 6 weighted dimensions: keyword match (30%), format parse-ability (20%), content quality (20%), section completeness (15%), file/format compliance (10%), contact completeness (5%). Return composite score 0–100.

FR-ATS-002: Two scoring modes: (a) Generic Role Scoring against built-in role keyword corpus; (b) JD-Specific Scoring — user pastes JD, system extracts required keywords and scores against them.

FR-ATS-003: For each dimension: sub-score + plain-English explanation of deductions + actionable recommendations.

FR-ATS-004: Upload support: PDF, DOCX, DOC. Multi-layer extraction (direct extraction → OCR fallback for image-based PDFs).

FR-ATS-005: Keyword gap analysis table: JD keywords, found in resume, missing high-priority, missing medium-priority, suggested phrasing.

FR-ATS-006: Simulate scoring behavior of: Taleo, Workday, Greenhouse, Lever, iCIMS, SAP SuccessFactors, Naukri RMS, PeopleStrong, Darwinbox, SmartRecruiters.

FR-ATS-007: Radar/spider chart visualization of composite score across all dimensions.

FR-ATS-008: Score history per resume for improvement tracking.

5.4 Module 4 — Job Intelligence Layer

FR-JI-001: Licensed API integrations: Adzuna, JSearch (RapidAPI), Jooble, Reed, Apna API. Zero scraping.

FR-JI-002: Search filters: job title, company, location (city/state/country/remote), employment type, experience level, salary range, date posted.

FR-JI-003: Job Compatibility Score (0–100): skill overlap (45%) + keyword similarity (25%) + experience match (20%) + location match (10%).

FR-JI-004: Coverage: local part-time/gig, domestic full-time, remote-eligible global, international MNC.

FR-JI-005: Job Alerts: email + in-app notifications for matching new listings.

FR-JI-006: Per-listing view: compatibility score, salary estimate, skills gap, apply link, "Tailor My Resume" button.

FR-JI-007: No scraping of any kind. All data via licensed APIs with proper attribution.

6. Software Requirements Specification (SRS)

6.1 System Overview

CareerOS is a web-first, mobile-responsive SaaS application: React/Next.js frontend, Python/FastAPI backend, PostgreSQL primary database, Redis for caching and queuing, and a dedicated AI inference layer. The ATS Engine microservice uses Intel OpenVINO Runtime for CPU-optimized NLP inference. The Job Intelligence service uses Intel scikit-learn-intelex (oneAPI AI Analytics Toolkit) for accelerated TF-IDF and cosine similarity operations. The full pipeline is developed and benchmarked on Intel DevCloud. The job intelligence layer integrates with external APIs through a licensed data aggregation service.

6.2 Performance Requirements

ID	Requirement	Target
NFR-PERF-001	Resume generation (profile to PDF)	P95 < 8 seconds
NFR-PERF-002	ATS scoring of uploaded resume	P95 < 12 seconds
NFR-PERF-003	Job search results return	P95 < 3 seconds
NFR-PERF-004	Concurrent user support	10K Phase 1 → 500K Phase 3
NFR-AVAIL-001	System uptime SLA	99.9% (max 8.7 hrs/year downtime)
NFR-AVAIL-002	Maintenance windows	Sun 2–4 AM IST, 48hr notice

6.3 Security Requirements

ID	Requirement
NFR-SEC-001	All user data at rest: AES-256 encryption
NFR-SEC-002	All data in transit: TLS 1.3
NFR-SEC-003	Passwords: bcrypt, min cost factor 12
NFR-SEC-004	OWASP Top 10 protections at launch
NFR-SEC-005	Resume files: per-user signed S3 URLs, 1-hour expiry
NFR-SEC-006	No hardcoded credentials; AWS Secrets Manager for all secrets
NFR-SEC-007	WAF on API Gateway, CloudTrail enabled

6.4 Compliance Requirements

DPDP Act 2023 (India): Explicit consent per data category at signup; data deletion within 30 days of account deletion request; data localization (Indian user data in AWS ap-south-1 / Mumbai); grievance mechanism in-app and via email.

GDPR (EU/UK): Lawful basis documented (Contractual Necessity + Legitimate Interest + Consent); data subject rights: access, rectification, erasure, portability, restriction, objection; DPA contacts and Article 30 records maintained.

6.5 External Interfaces

Interface	Provider	Purpose	Protocol
AI LLM APIs	Anthropic Claude (primary), GPT-4o, Gemini 1.5 Flash	Resume generation, content enhancement	HTTPS REST, API Key (server-side only)
Job Data APIs	Adzuna, JSearch, Jooble, Reed, Apna	Licensed job listings aggregation	HTTPS REST, various auth
Authentication	JWT + Google OAuth2 + LinkedIn OAuth2	User auth + LinkedIn import	OAuth2 / JWT
Email Service	AWS SES (primary), Resend (fallback)	Transactional email, job alerts	SMTP / REST API
File Storage	AWS S3 (ap-south-1)	Resume files, template assets	AWS SDK / Signed URLs
Payment	Razorpay (India), Stripe (international)	Subscription billing	HTTPS REST, Webhooks
Intel AI Toolkit (EI-007)	OpenVINO Runtime, OpenVINO Model Optimizer, PEX, Intel oneAPI Deep Learning SDK	On-premise inference processing on dedicated hardware	OpenVINO SDK

7. System Architecture

7.1 Architecture Overview

CareerOS uses a microservices architecture deployed on AWS. Four independently deployable services communicate through a shared API Gateway. The data layer is shared (PostgreSQL + Redis + S3) with row-level access controls between services. Intel technologies are integrated at two critical microservice layers: OpenVINO Runtime in the ATS Engine, and scikit-learn-intelex in the Job Intelligence service.

Layer	Technology	Key Responsibilities
Client	Next.js (React), Tailwind CSS	Web app, mobile-responsive, SSR for SEO
API Gateway	AWS API Gateway + Lambda@Edge	Rate limiting, auth, SSL termination, routing
Core API	Python 3.11, FastAPI, SQLAlchemy	User/profile/resume CRUD, billing, auth
ATS Engine	Python 3.11, FastAPI, spaCy, pdfplumber, Intel OpenVINO Runtime, OpenBLAS	Resume parsing, INCO Profile generation (OpenVINO), scoring, recommendations
Job Intelligence	Python 3.11, FastAPI, APScheduler, Intel scikit-learn-intelex	API aggregation/indexing, compatibility scoring (11-12x faster via Intel oneAPI)
AI Inference	Python 3.11, FastAPI, Anthropic/OpenAI/Gemini SDKs	CV/GenImage SDXLs, routing, fallback
Primary DB	PostgreSQL 15 (AWS RDS Multi-AZ)	All persistent structured data
Cache/Queue	Redis 7 (AWS ElastiCache)	Session cache, Celery task queue, dedup
File Storage	AWS S3 (ap-south-1)	Resumes, exports, templates
Dev & Benchmarking	Intel DevCloud	OpenVINO vs. baseline benchmarks, latency profiling, SLA validation

7.2 Scalability Design

- All application tier services are stateless — horizontal scaling behind an Application Load Balancer.
- ATS Engine deployed on Intel Xeon compute-optimized instances (c6i.2xlarge) and auto-scales based on Celery queue depth.
- PostgreSQL uses read replicas for analytics queries; primary handles writes only.
- Redis cluster mode for cache scalability; separate queue and cache instances in production.
- S3 for all binary storage — no file system dependency, infinite scale.
- CDN (CloudFront) for static assets and template previews — reduces origin load.

7.3 Intel Technology Stack — Full Integration Map

Intel technologies are integrated across the full AI project lifecycle — from development and benchmarking through to production deployment:

Intel Technology	Component	Lifecycle Stage	Measured Impact
Intel OpenVINO Runtime	ATS Engine — spaCy NER Model	Model Deployment	3-4x latency reduction: 420ms -> 110ms per resume parse
OpenVINO Model Optimizer	ATS Engine — ONNX to IR	Evaluation Deployment	Compiles spaCy transformer to IR; one-time build step
Intel Extension for PyTorch (IPEX)	ATS Engine — PyTorch pre-modeling	Deployment	15-25% throughput on tokenization via operator fusion
Intel scikit-learn-intelex (oneAPI)	Job Intelligence — TF-IDF Model	Model Deployment	11-12x speedup; enables 3s P95 SLA on CPU-only infra
Intel DevCloud	Full ATS + Job Intelligence	Development + Evaluation	Free Xeon sandbox for benchmarking, profiling, SLA validation
Intel Xeon Scalable (c6i)	ATS + Job Intelligence mid-layer	Deployment	AVX-512 instruction set fully exploited by sklearnex

Economic rationale: At Rs.299/month Pro pricing with 30 ATS scans included, each scan has a maximum viable compute budget of ~Rs.0.30. Intel OpenVINO + scikit-learn-intelex enables CPU-only deployment that meets all latency SLAs within this budget. GPU infrastructure would make this price point economically impossible — Intel optimization is directly load-bearing for the inclusion mission of the product.

8. AI & Machine Learning Design

8.1 Resume Content Generation

Task: Given a structured career profile and target role, generate professionally worded resume content.

Approach: Prompt engineering on Claude claude-sonnet with structured JSON input and constrained JSON output. No fine-tuning required in Phase 1.

The AI is instructed as a senior technical recruiter who has reviewed 10,000+ resumes. It understands STAR format, ATS keyword density, and the difference between Indian and global resume conventions. It generates content grounded in the user's actual data — it never fabricates experience, certifications, or metrics.

Safety guardrail: The AI shall not generate false claims, fabricated companies, invented certifications, or inflated metrics. All numeric claims are validated against user-provided data.

8.2 ATS Scoring Model

Phase 1 approach: Rule-based weighted scoring system, informed by published ATS research papers and reverse-engineering documentation of major ATS systems. This is deterministic, not ML-based.

Why rule-based first: Explainability is a core product requirement. Users need to understand exactly why their score is what it is. Phase 1 prioritizes trust and actionability over marginal accuracy gains.

Intel OpenVINO Integration: The underlying NLP pipeline (section detection, entity extraction, keyword identification) uses a spaCy transformer model running on Intel OpenVINO Runtime. Exported from PyTorch via ONNX then compiled with `ov.Core().compile_model()` — reducing NLP inference from ~420ms (PyTorch CPU baseline) to ~110ms (OpenVINO on Intel Xeon). This is what makes the P95 < 12s ATS scan SLA achievable without GPU infrastructure.

Dimension	Weight	Rationale
Keyword match (role-based or JD)	30%	Highest impact on ATS filtering decisions
Section parse-ability	20%	Format failures are most common silent rejection cause
Content quality (STAR, quantification)	20%	Quantified bullets score higher in AI-screening ATS
Section completeness	15%	Missing sections cause data loss in parsing
File/format compliance	10%	Wrong file type, oversized files trigger auto-rejection
Contact info completeness	5%	Incomplete contacts = unparseable application

Phase 2 ML upgrade: Once 100,000+ ATS scans are completed, an XGBoost classifier will be trained on outcome data. It will be exported to OpenVINO IR format via `openvino.convert_model()` — using the same

OpenVINO Runtime pipeline already deployed, with zero infrastructure changes.

8.3 Job Compatibility Scoring Algorithm

Intel scikit-learn-intelex (oneAPI): All TF-IDF and cosine similarity operations use Intel scikit-learn-intelex — patched with a single line at startup. Benchmarks show 11-12x speedup (TF-IDF: 4.2s -> 0.38s; cosine similarity: 890ms -> 71ms on Intel Xeon). Required to meet the 3-second job search P95 SLA at scale.

Formula: $\text{compatibility} = (0.45 \times \text{skill_overlap}) + (0.25 \times \text{keyword_similarity [Intel sklearnex]}) + (0.20 \times \text{experience_match}) + (0.10 \times \text{location_match})$

- Skill overlap: Jaccard similarity between extracted skills from resume vs. JD required skills.
- Keyword similarity: Cosine similarity on TF-IDF vectors — accelerated by Intel scikit-learn-intelex (oneAPI).
- Experience match: Gaussian decay — exact match = 1.0, +/-1 year = 0.85, +/-2 years = 0.65, >3 years off = 0.3.
- Location match: Exact city = 1.0, same state = 0.8, remote-eligible = 0.9, different country = 0.2.

9. Database Design

9.1 Core Tables

users

```
id (UUID PK), email (unique), password_hash, full_name, phone, city, country, linkedin_url,
github_url, professional_status, created_at, updated_at, subscription_tier (free/pro/premium),
subscription_expires_at, is_verified, is_active, last_login_at
```

career_profiles

```
id (UUID PK), user_id (FK), target_role, target_industry, location_preference,
professional_summary, years_of_experience, profile_completeness_score (0-100), created_at,
updated_at
```

work_experiences

```
id (UUID PK), career_profile_id (FK), company_name, job_title, employment_type, location,
start_date, end_date, is_current, description_bullets (JSONB), created_at, updated_at
```

educations

```
id (UUID PK), career_profile_id (FK), institution_name, degree_type, field_of_study,
start_year, end_year, cgpa_or_percentage, relevant_coursework (JSONB)
```

skills

```
id (UUID PK), career_profile_id (FK), skill_name, category (technical/soft/language/tool),
proficiency_level
```

projects

```
id (UUID PK), career_profile_id (FK), title, description, tech_stack (JSONB), github_url,
live_url, start_date, end_date
```

certifications

```
id (UUID PK), career_profile_id (FK), name, issuing_organization, issue_date, expiry_date,
credential_id, credential_url
```

resumes

```
id (UUID PK), user_id (FK), career_profile_id (FK), name, template_id, custom_section_order
(JSONB), custom_overrides (JSONB), s3_key, last_generated_at, created_at, updated_at
```

ats_scans

id (UUID PK), user_id (FK), resume_id (FK nullable), source_type, uploaded_file_s3_key, jd_text, composite_score, dimension_scores (JSONB), recommendations (JSONB), keyword_gaps (JSONB), scanned_at

job_listings

id (UUID PK), external_id, source_api, title, company, location, country, remote_eligible, employment_type, experience_level, salary_min, salary_max, description_text, required_skills (JSONB), extracted_keywords (JSONB), apply_url, posted_at, fetched_at, expires_at

job_compatibility_scores

id (UUID PK), user_id (FK), job_listing_id (FK), composite_score, skill_overlap_score, keyword_score, experience_match_score, location_match_score, computed_at

9.2 Key Indexes

Table	Index Column(s)	Type	Purpose
users	email	Unique B-tree	Auth lookup
career_profiles	user_id	B-tree	Profile retrieval
job_listings	posted_at	B-tree	Recency sorting
job_listings	location, employment_type, experience_level	Composite B-tree	Filtered search
job_compatibility_scores	user_id, composite_score	Composite B-tree	Ranked job matches
ats_scans	user_id, scanned_at	Composite B-tree	Scan history queries

10. API Design

10.1 Conventions

- Base URL: `https://api.careeros.in/v1`
- Protocol: REST over HTTPS, JSON request/response
- Auth: Bearer JWT in Authorization header
- Rate limits: Free — 100 req/hr; Pro — 1,000 req/hr; Premium — 10,000 req/hr
- Versioning: URL path (`/v1/`, `/v2/`)
- Error format: `{ "error": { "code": "RESOURCE_NOT_FOUND", "message": "...", "details": {} } }`

10.2 Key Endpoints

Authentication

Method	Endpoint	Description
POST	<code>/auth/register</code>	Create account
POST	<code>/auth/login</code>	Get JWT token
POST	<code>/auth/refresh</code>	Refresh token
POST	<code>/auth/google</code>	Google OAuth callback
POST	<code>/auth/linkedin</code>	LinkedIn OAuth + profile import
POST	<code>/auth/forgot-password</code>	Trigger password reset email

Career Profile

Method	Endpoint	Description
GET	<code>/profile</code>	Get my career profile
PUT	<code>/profile</code>	Update career profile
POST	<code>/profile/work-experience</code>	Add work experience entry
PUT	<code>/profile/work-experience/:id</code>	Update entry
DELETE	<code>/profile/work-experience/:id</code>	Remove entry
POST	<code>/profile/skills/bulk</code>	Bulk upsert skills

Method	Endpoint	Description
POST	/profile/import-linkedin	Import from LinkedIn
GET	/profile/completeness	Get completeness score

Resume Builder

Method	Endpoint	Description
GET	/resumes	List my resumes
POST	/resumes	Create new resume
PUT	/resumes/:id	Update resume settings
POST	/resumes/:id/generate	Generate PDF (returns signed S3 URL)
GET	/resumes/:id/preview	Get HTML preview
POST	/resumes/:id/duplicate	Clone resume
GET	/templates	List available templates

ATS Engine

Method	Endpoint	Description
POST	/ats/scan/resume/:id	Score a CareerOS-built resume
POST	/ats/scan/upload	Upload + score external resume (multipart/form-data)
POST	/ats/scan/jd-match	Score resume against pasted JD
GET	/ats/scans	Get scan history
GET	/ats/scans/:id	Get detailed scan report
GET	/ats/benchmarks/:role	Get benchmark scores for a role

Job Intelligence

Method	Endpoint	Description
GET	/jobs	Search jobs (q, location, type, level, remote)
GET	/jobs/matches	Get personalized matches for my profile
GET	/jobs/:id/compatibility	Get compatibility score for a job
POST	/jobs/alerts	Create a job alert

Method	Endpoint	Description
GET	/analytics/dashboard	Career health overview
GET	/analytics/market	Market intelligence for my role

11. UI/UX Flow Design

11.1 Onboarding Flow

- Landing Page → Sign Up (email / Google)
- Email Verification → Professional Status Selection
- (Student path) College + Degree → Target Role + Industry
- Skills Entry (tag-based) → Projects Entry (1 minimum nudge)
- Profile Completeness Score shown (e.g., 62%)
- CTA: "Build Your First Resume" or "Score Your Existing Resume"

11.2 Resume Builder Flow

- Select Mode: "Start from my profile" or "Start fresh"
- Choose Template (8 shown with ATS compatibility badge)
- AI-Guided Sections: Personal Info → Summary → Work Experience → Education → Skills → Projects → Certifications → Optional Sections
- Live Preview (side-by-side desktop / toggle mobile)
- ATS Pre-Check (automatic quick score before download)
- Download PDF / Save to My Resumes

11.3 ATS Scoring Flow

- Choose Input: "Score my CareerOS resume" or "Upload existing resume"
- (Optional) Paste Job Description for JD-specific scoring
- Processing animation (~8–12 seconds)
- Results: Composite Score (0-100) + Radar Chart
- Dimension Cards: each with sub-score, explanation, fix instructions
- Keyword Gap Analysis table + Prioritized Fix-It List
- CTA: "Fix These Issues" → opens resume editor

11.4 Job Search Flow

- Jobs Home: Personalized Matches (auto-loaded) + Search + Filters
- Job Cards: Title, Company, Location, Compatibility Score %, Salary, Date
- Quick Actions per card: View | Tailor Resume | Track Application
- Job Detail: Full JD + Compatibility breakdown + Skills gap

- "Tailor My Resume" → generates role-specific resume version
- "Apply Now" → opens source URL in new tab

12. ATS Engine Specification

12.1 Resume Parsing Pipeline

Stage 1 — File Processing

Accept PDF and DOCX. For PDF: attempt direct text extraction via pdfplumber. If extracted text is <100 characters (image-based PDF), invoke Tesseract OCR pipeline. For DOCX: extract via python-docx preserving paragraph structure. Output: raw text + paragraph list + detected layout flags.

Stage 2 — Section Identification

Apply trained spaCy NER model + rule-based regex to identify sections: Contact Info, Summary, Work Experience, Education, Skills, Projects, Certifications, Awards, Languages, Publications, Volunteer Work. Flag unclassified content.

Stage 3 — Entity Extraction

Contact Info: name, email, phone, city, URLs. Work Experience: company, title, dates, bullets. Education: institution, degree, dates, CGPA. Skills: individual tokens. Output: structured data object.

Stage 4 — Multi-System Scoring

Run all 6 scoring dimensions (keyword match, format, content quality, completeness, file compliance, contact completeness). Each ATS system simulation applies different weights. Scores aggregated by market share weights into composite score.

12.2 ATS System Simulation Weights

ATS System	Market Weight (India+Global)	Behavioral Notes
Taleo (Oracle)	18%	Very strict on section headers; penalizes tables and text boxes heavily
Workday	16%	Strong keyword matching; good at parsing modern single-column formats
Naukri RMS	15%	India-specific; handles Indian degree names and company names well
Greenhouse	12%	Lenient on format; strong keyword density scoring; startup-friendly
PeopleStrong	10%	Indian HR tech platform; handles regional name variations
Darwinbox	9%	Fast-growing in India; similar parsing behavior to Workday
Lever	8%	Startup-focused; modern parser; reasonable 2-column support
iCIMS	7%	Enterprise US/EU; very strict formatting; penalizes unusual section names

ATS System	Market Weight (India+Global)	Behavioral Notes
SmartRecruiters	5%	Global platform; lenient parser; good at automated skills extraction

13. Job Intelligence Engine

13.1 Data Sources and Coverage

API	Coverage	Daily Listings	Strength
Adzuna	Global (16 countries + India)	500K+	Best India + global balance; long-standing licensed API
JSearch (RapidAPI)	Global aggregator	1M+	Real-time; broadest coverage; single API for multiple sources
Jooble	71 countries	200K+	Strong Europe + emerging markets coverage
Reed	UK + some global	100K+	International premium roles; reliable data quality
Apna API	India (tier-2/3 focus)	50K+	Entry-level, blue-collar, local jobs — critical for our core user

13.2 Deduplication Logic

Jobs are considered duplicates if: company name similarity >90% AND job title similarity >85% AND location exact match AND posted within 48 hours. Deduplication uses MinHash LSH (Locality Sensitive Hashing) for scalable approximate matching at high volume.

13.3 Job Categorization

- Tier:** Local Part-time / Gig | National Full-time | Remote Global | MNC International
- Level:** Internship | Entry (0–2 yrs) | Mid (2–5 yrs) | Senior (5–10 yrs) | Lead/Manager | Executive
- Domain:** Technology | Finance | Marketing | Operations | Design | Healthcare | Education | Manufacturing | Other
- Work Mode:** On-site | Remote | Hybrid

14. Security & Compliance

14.1 Authentication & Authorization

- JWT access tokens: 15-minute expiry.
- Refresh tokens: 30-day expiry, stored in httpOnly cookie (never localStorage).
- Role-based access control: User | CollegeAdmin | Recruiter | PlatformAdmin.
- All admin actions logged: user ID, timestamp, action type, IP address.
- MFA support: TOTP-based (Google Authenticator compatible) for Pro and Premium users.

14.2 DPDP Act 2023 (India) Compliance

- Consent Management: explicit opt-in for each data category at signup.
- Data Principal Rights: access my data (JSON export), correct, erase (30-day SLA), grievance mechanism.
- Purpose limitation and storage limitation enforced at application level.
- Data localization: all Indian user data stored in AWS ap-south-1 (Mumbai).
- Consent audit logs maintained for regulatory inspection.

14.3 GDPR Compliance

- Lawful basis documented: Contractual Necessity + Legitimate Interest + Consent.
- Data Subject Rights implemented: access, rectification, erasure, portability, restriction, objection.
- DPA contacts appointed; Article 30 Records of Processing maintained.
- Privacy by design: minimum data collection; data not used beyond declared purposes.

14.4 Resume Data Handling

Uploaded resumes are processed in memory where possible. Temporary files are written to an encrypted S3 prefix and deleted within 1 hour of processing. Users explicitly choosing to save uploaded resumes receive clear notice. Resume data is never sold, shared with third parties, or used for any purpose other than providing CareerOS services without explicit user consent.

15. Phased Roadmap

Phase 1 — Foundation (Weeks 1–10)

Goal: Shippable, monetizable core product. Resume builder + ATS engine only. Each phase is independently monetizable.

Week	Milestone
1–2	Project setup, tech stack, DB schema, auth system, career profile CRUD. Set up Intel DevCloud workspace; establish NLP in
3–4	Resume template engine, HTML/CSS templates (4 designs), PDF generation via WeasyPrint.
5–6	AI resume builder integration (Claude API), content generation, professional summary. Apply Intel IPEX to PyTorch preprocess
7–8	ATS engine v1 (rule-based scoring). Export spaCy NER to ONNX → OpenVINO IR. Benchmark OpenVINO vs. PyTorch CPU
9	Integrate Intel scikit-learn-intelex into Job Intelligence scoring. UI polish, mobile responsiveness, Razorpay payment integratio
10	Closed beta (500 users), feedback, bug fixes. Final Intel DevCloud performance report documenting OpenVINO and sklearn

Phase 2 — Intelligence (Weeks 11–22)

Week	Milestone
11–13	Job API integrations (Adzuna, JSearch, Jooble), aggregation + dedup pipeline
14–15	Compatibility scoring algorithm, job match UI, job detail pages
16–17	Job alerts (email + in-app), application tracker, resume version history
18–19	ATS engine v2 (4 additional ATS systems), accuracy improvements
20–21	Analytics expansion, market intelligence, career health dashboard
22	Phase 2 public launch, scaling test, press/marketing push

Phase 3 — Scale (Months 6–12)

- B2B college portal: placement officer dashboard, bulk student management, placement analytics.
- Recruiter access module: search resume database, shortlisting tools, subscription billing.
- Mobile app: React Native (Android + iOS).
- Cover letter generator + LinkedIn profile optimizer.
- ATS engine ML upgrade: train on collected scan data.
- International expansion: Southeast Asia (Philippines, Indonesia, Vietnam).

- Series A fundraising preparation.

16. Monetization & Business Model

16.1 Individual (B2C) Pricing Tiers

Feature	Free	Pro (₹299/mo)	Premium (₹799/mo)
Resume builds/month	3	Unlimited	Unlimited
ATS scans/month	5	30	Unlimited
Job search queries/day	3	Unlimited	Unlimited
Templates available	2	8+	8+ (incl. future)
JD-specific scoring	No	Yes	Yes
PDF export (no watermark)	No	Yes	Yes
Resume versions	1	10	Unlimited
Job alerts	No	3 active	Unlimited
Cover letter (Phase 2)	No	No	Yes
LinkedIn optimizer (Phase 3)	No	No	Yes
Priority support	No	No	Yes

16.2 Institutional (B2B) Pricing

Plan	Price/Year	Capacity	Key Features
College Starter	₹49,999	Up to 500 students	Placement dashboard, bulk resume scoring, placement analytics
College Pro	₹1,49,999	Up to 2,000 students	Starter + custom branding, API access, performance reports, account management
Enterprise	Custom	Unlimited	Pro + white-label option, SLA guarantees, on-premise deployment, custom integrations

16.3 Recruiter Access

Pay-per-search: ₹199 per batch of 100 profiles. Monthly subscription: ₹4,999/month for unlimited filtered searches. Recruiters access CareerOS's verified, ATS-scored resume database. User participation is opt-in — users choose whether their profile is visible to recruiters.

16.4 Revenue Projections (Conservative)

Quarter	B2C MRR	B2B MRR	Recruiter MRR	Total MRR
Q1 (post-launch)	■1.2L	—	—	■1.2L
Q2	■4.5L	■2.0L	■0.5L	■7.0L
Q3	■12L	■8L	■2L	■22L
Q4	■28L	■18L	■6L	■52L
Year 2 ARR (target)	—	—	—	■8–10 Cr

17. Risk Register

Risk	Prob.	Impact	Mitigation
AI API cost overrun	Medium	High	Caching, use Haiku for lightweight tasks, hard spending caps, multi-provider routing
Job API rate limits or pricing changes	Medium	Medium	Multi-API strategy, caching, contractual SLAs, minimum 3 active providers
Competitor copies ATS engine	Medium	Medium	Speed of execution + data moat is defense; file patents for scoring methodology
DPDP compliance gap found post-launch	Low	Very High	Engage compliance counsel pre-launch, regular audits, privacy-by-design from start
PDF generation quality / ATS compatibility issues	High	High	Extensive testing across 50+ ATS systems pre-launch; maintain fallback templates
Low freemium-to-paid conversion	Medium	High	Aggressive usage limits on free tier, clear value demonstration before paywall hit
Single AI provider dependency	Low	Very High	Multi-provider fallback (Claude → GPT-4o → Gemini) already in architecture
College B2B sales cycle too long	High	Medium	Start with small colleges (500 students), offer 3-month free pilots, track placement
Scaling infrastructure costs exceed revenue	Low	Medium	Horizontal scaling architecture, query optimization, CDN, aggressive caching
Key engineering talent leaving	Medium	Medium	Competitive equity compensation, meaningful technical ownership, documentation

18. Glossary

ATS (Applicant Tracking System)

Software used by employers to manage and filter job applications. Examples: Taleo (Oracle), Workday, Greenhouse, Naukri RMS.

ATS Score

CareerOS's proprietary measure (0–100) of how well a resume will perform when processed by ATS software.

Career Profile

The persistent, structured data store in CareerOS that holds all of a user's career information and serves as the single source of truth for all features.

Compatibility Score

CareerOS's measure (0–100) of how well a user's profile matches a specific job listing.

DPDP Act

India's Digital Personal Data Protection Act, 2023. Governs how personal data of Indian citizens must be handled by data fiduciaries.

JD (Job Description)

The text posted by an employer describing a job opening, including responsibilities, required qualifications, and skills.

Job Intelligence Layer

CareerOS's module that aggregates live job listings from licensed APIs and performs compatibility matching against user profiles.

Keyword Gap Analysis

Identification of keywords present in a JD but absent from a resume, with suggestions for natural incorporation.

MinHash LSH

MinHash Locality Sensitive Hashing — a probabilistic algorithm used for scalable approximate deduplication of job listings.

NER (Named Entity Recognition)

An NLP technique used in CareerOS's ATS engine to identify and classify entities (companies, titles, dates, skills) in resume text.

STAR Format

Resume writing framework — Situation, Task, Action, Result — for structuring achievement-oriented bullet points.

TF-IDF

Term Frequency–Inverse Document Frequency — a numerical statistic used in CareerOS job matching to assess keyword relevance.

WeasyPrint

Python library used to generate PDF files from HTML/CSS templates in CareerOS, producing ATS-safe, text-selectable output.

spaCy

Industrial-strength NLP library used in CareerOS's ATS engine for section detection, entity extraction, and text analysis. Its transformer model backend runs on Intel OpenVINO Runtime in production.

Intel OpenVINO Runtime

Intel's open-source toolkit for optimizing and deploying AI inference on Intel hardware. Used in CareerOS's ATS Engine to run the spaCy NER model on Intel Xeon CPUs without a GPU, achieving 3-4x latency improvement over standard PyTorch CPU inference.

OpenVINO Model Optimizer

Tool that converts trained models (PyTorch, ONNX, TensorFlow) to OpenVINO Intermediate Representation (IR) format for optimized Intel CPU deployment. Used to convert CareerOS spaCy model: PyTorch -> ONNX -> OpenVINO IR.

Intel Extension for PyTorch (IPEX)

Intel's optimization library for PyTorch that improves performance on Intel CPUs via operator fusion and memory layout optimization. Applied to CareerOS's NLP preprocessing pipeline with a single `ipex.optimize()` call.

Intel scikit-learn-intelex

Part of the Intel oneAPI AI Analytics Toolkit. Patches scikit-learn to run on Intel hardware with 11-12x speedup on TF-IDF and cosine similarity — critical for CareerOS's job compatibility scoring at scale.

Intel DevCloud

Intel's free cloud development sandbox providing access to Intel Xeon Scalable processors and AI accelerators. Used by CareerOS for benchmarking OpenVINO vs. baseline inference, profiling memory, and validating latency SLAs before production deployment.

oneAPI

Intel's unified programming model for cross-architecture development. CareerOS uses the oneAPI AI Analytics Toolkit component (scikit-learn-intelex) for accelerated machine learning operations.

Building careers, not just resumes.

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